

Akshat G

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Research Interests

Data-centric and data-efficient machine learning, self-supervised representation learning, and generative pretraining, with an emphasis on structured and hierarchical prediction in low-label regimes. Research experience spans diffusion- and GAN-based pretraining for medical image segmentation, semi-supervised learning under extreme label scarcity, and hierarchy-aware architectures for structured visual prediction.

Education

Manipal Institute of Technology

Bachelor of Technology (Honours) — Computer Science and Engineering

Bengaluru, India

Jul 2023 – May 2027

- **Specialization:** Artificial Intelligence **Minor:** AI in Healthcare
- **CGPA:** 9.56/10 **Awards:** 2× Achievers Scholarship (Top 5%)

Research Experience

HiME: Hierarchical Multi-Label Vision Architecture

Shell India Markets Pvt. Ltd. — Intern, Multiphysics AI & Product Innovation
Advisor — [Vishal Ahuja](#)

Bengaluru, India

Jun 2026 – Present

- Designed **HiME**, a hierarchical multi-label vision architecture that enforces hierarchical label consistency **by construction** rather than through post-hoc correction, applied to gear failure analysis on industrial imaging data.
- Achieved **100% hierarchy consistency** and **90% expert-validated accuracy**, outperforming post-processing-based hierarchical correction baselines.
- Built a modular inference framework decoupling feature extraction from hierarchical classification, enabling rapid benchmarking across multiple vision backbones.

Unsupervised Anatomical Feature Learning via Diffusion Models

Manipal Institute of Technology

First Author — Advisors: [Dr. Shaleen Bhatnagar](#), [Dr. Shilpa Ankalaki](#), [Dr. Tusar Kanti Misra](#)

Bengaluru, India

Feb 2026 – Apr 2026

- Proposed using unsupervised Denoising Diffusion Probabilistic Models (DDPMs) as a pretraining strategy for medical image segmentation, training a DDPM on unlabeled abdominal CT scans (BTCV dataset) and transferring the learned encoder to downstream segmentation.
- Demonstrated statistically significant gains from diffusion pretraining (Dice: liver $0.75 \rightarrow 0.93$, $p < 10^{-25}$; kidney $0.90 \rightarrow 0.95$, $p < 10^{-10}$), with 45–66% reductions in boundary error (HD95, ASD).
- Showed frozen, never-fine-tuned diffusion encoders retain **>80% of fully fine-tuned performance** on liver segmentation, providing direct evidence that unsupervised pretraining learns transferable anatomical priors rather than generic features.
- Validated robustness under label scarcity: diffusion-pretrained models retained 95.5% of full-data liver performance using only 10% of labels.
- *Submitted to IEEE Access; under review.*

Quality-Controlled Synthetic Medical Data Generation

Manipal Institute of Technology — Honors Thesis

Solo Lead Researcher — Advisor: [Dr. Rashmi R](#)

Bengaluru, India

Aug 2025 – Jan 2026

- Developed a Teacher–Student semi-supervised framework for lumbar spine MRI segmentation (SPIDER dataset) to address extreme label scarcity.
- Engineered a conditional GAN (cGAN) to synthesize pseudo-images and proposed a novel quantitative criterion for pseudo-label quality assessment, filtering GAN-generated noise.
- Improved student-model performance by **4% (Dice Score)** over the Teacher baseline.
- *Submitted; under review.*

Data-Centric Polyp Segmentation for Colorectal Cancer Detection

Manipal Institute of Technology

Solo Lead Researcher — Advisor: [Dr. Rashmi R](#)

Bengaluru, India

Aug 2025 – Oct 2025

- Developed a pre-processing pipeline with enhanced ground-truth labels and a novel deep learning framework for polyp segmentation in colonoscopy images.
- Achieved **IoU of 0.9115** and **Dice score of 0.9525** on held-out test sets, outperforming existing literature benchmarks.
- *Submitted; under review.*

Nucleus Segmentation in H&E-Stained Histopathology Images

Manipal Institute of Technology

First Author — Advisor: [Dr. Tusar Kanti Misra](#)

Bengaluru, India

Aug 2025 – Oct 2025

- Conducted experimental analysis on H&E-stained images using the MoNuSeg dataset, achieving a competitive Dice score of ~ 0.80 relative to current SOTA architectures.
- *Accepted, SAGE Digital Health; awaiting publication.*

Semi-Supervised Learning on Constrained Datasets

Manipal Institute of Technology

First Author — Advisors: [Dr. Rashmi R](#), [Dr. Girija Surathkal](#)

Bengaluru, India

May 2025 – Aug 2025

- Conducted experimental analysis of teacher–student training dynamics using only 50 labeled samples.
- Benchmarked Attention U-Net against SOTA architectures (U-Net++, DeepLabV3, DINO) and designed an adaptive confidence-thresholding strategy based on pixel-wise uncertainty.
- Demonstrated a **10% relative improvement** in segmentation accuracy under low-label conditions.
- *Submitted; under review.*

Segmenting Lung Regions for TB Detection

Manipal Institute of Technology

First Author — Advisor: [Dr. Rashmi R](#)

Bengaluru, India

Apr 2025 – May 2025

- Benchmarked U-Net and advanced variants for region-of-interest segmentation in chest X-rays, optimizing hyperparameters to achieve a **Mean Dice of 0.95**.
- *Presented at CVIP 2025; awaiting publication.*

Publications

- Mishra, N., **Akshat, G.**, Moorthy, U., Sathishkumar, V. E. *Binary Classification of Galaxies Using Machine Learning and Deep Learning*. In: *Data Science and Exploration in Artificial Intelligence* (CODE-AI 2025), Communications in Computer and Information Science, vol. 2689, Springer, Cham, 2026, pp. 88–98. [doi:10.1007/978-3-032-19318-6_9](#)
- **Akshat, G.**, Misra, T. K. *Nucleus Segmentation in H&E-Stained Histopathology Images*. **Accepted**, SAGE Digital Health; awaiting publication.

Manuscripts under review (first author unless noted)

- *Unsupervised Anatomical Feature Learning via Diffusion Models: Enhanced Medical Image Segmentation with DDPMs*. Submitted to IEEE Access.
- *Quality-Controlled Synthetic Medical Data Generation* (Honors Thesis). Submitted.
- *Semi-Supervised Learning on Constrained Datasets*. Submitted.
- *Data-Centric Polyp Segmentation for Colorectal Cancer Detection*. Submitted.

Conference Presentations (first author)

- *Segmenting Lung Regions for TB Detection*. Presented at CVIP 2025; awaiting publication.

Technical Skills

- **Languages:** Python, C, C++, Java
- **ML/DL Frameworks:** PyTorch, TensorFlow, Keras, HuggingFace Transformers, scikit-learn
- **Core Methods:** Diffusion Models (DDPMs), GANs, Semi-/Self-Supervised Learning, Transfer Learning, Hierarchical & Structured Prediction, Segmentation (U-Net, U-Net++, Attention U-Net, DeepLabV3), Transformers